Shared Subtree Budgets and Quantitative Descent for Countably Branching Diamonds

Dr. Muralidhara Subbarao
Technical Report Assisted by AI Models
www.integralresearch.net
Stony Brook University
Technical Report TR-Ram-OpenAI-722-M639

October 9, 2026

Abstract

The assigned manuscript derives diamond distortion from several different midpoint and tree-energy estimates. We refine its direct primal-test proof in the finite-height segment-forest dual. Grouping a crossing segment with the remaining tests in the same fresh subtree reduces the correction cost from sixteen to twelve. Retaining the initial norming test and removing avoidable selection slack gives $D^2 \geq 1 + k/12$ for depth k, independently of the companion midpoint theorem. This does not improve the source's stronger, midpoint-based $D^2 \geq 1 + k/4$. An adaptive finite-error witness budget and an explicit finite-head packing condition specify what a finite selection can certify, without asserting uniform finite-branch ranks. For the distinct fixed-zero-root recursive norm, the source's imported cubic input yields a retained-radius recurrence and the conditional bound $k \leq \frac{4}{3}((2D+1)^3 - 27)$, substantially reducing its displayed finite-error constant. Exact local inequalities, quantitative examples and the limits of finite verification are given. Contribution assessment: C3/5, shared subtree accounting, an adaptive descent certificate, and a retained-radius cubic transfer give notable source-specific refinements without a new global embedding classification.

Contents

1 Exact graph and norm conventions **2**

2 A shared fresh-subtree correction budget **3**

3 An adaptive witness and its initial budget **4**

4 What finite branching can actually certify **5**

5 Retaining the radius in a different cubic norm **6**

6 Explicit comparisons and certificate boundaries **8**

7 Literature, limitations and research directions **9**

8 Verification and contribution scope **10**

1 Exact graph and norm conventions

All vectors and Banach spaces below are real. The normalized diamond $\mathcal{D}_0$ has one edge of length one. At each replacement an edge $[a, b]$ of length ℓ is replaced by countably many paths $[a, m_i, b]$, with distinct new vertices and two edges of length $\ell/2$. The shortest-path metric on $\mathcal{D}_k$ is denoted d_k. Old-vertex distances do not change: a replacement preserves old path lengths, and compressing each two-edge portion of a new walk gives an old walk of no greater length. Consequently a stage-j edge has endpoint distance 2^{-j}, and two of its distinct new midpoints have distance 2^{-j}. This last distance, not just edge lengths, is essential.

A distortion-D embedding means a map f and a scalar $s > 0$ with

$$sd_k(u, v) \le \|f(u) - f(v)\| \le Dsd_k(u, v) \quad \text{for all vertices } u, v. \tag{1}$$

We divide by s unless otherwise stated. Then the lower factor is one, the upper factor is D, and $D \ge 1$. For an oriented parent define

$$x = \frac{f(b) - f(a)}{\ell}, \qquad z_i = \frac{2f(m_i) - f(a) - f(b)}{\ell}. \tag{2}$$

The two actual child slopes are $x + z_i, x - z_i$. Every slope has norm between one and D, every $\|z_i\| \le D$, and $\|z_i - z_l\| \ge 2$ for $i \ne l$. Choosing a sign after choosing i therefore follows an actual graph edge.

1.1 The finite-height forest

Let $T = \coprod_{h \ge 1} \mathbb{N}^{\le h}$, with a separate root in each component and the prefix order. A segment $[a, b]$ consists of all consecutive vertices between comparable a, b. For a finitely supported array v put

$$\rho(v) = \sup_{\mathcal{I}} \left(\sum_{I \in \mathcal{I}} \left| \sum_{t \in I} v(t) \right|^2 \right)^{1/2}, \tag{3}$$

where $\mathcal{I}$ ranges over finite families of pairwise set-disjoint segments. Complete in this norm to obtain J; the target of the direct argument is exactly $X = J^*$, with its dual norm. Disjoint segments need not be incomparable.

A set A is ancestral when it contains every ancestor of each of its vertices. Write P_A for coordinate restriction and $Q_A = I - P_A$. We will use finite ancestral A. Here both projections are contractive. The reason is that each admissible segment meets A, and also its complement, in a segment or the empty set; disjointness is preserved. The statement passes from J to X by duality.

For clarity, the finite-coordinate density needed in X does not follow just from the definition of a dual. On one height-h component,

$$\|v\|_2 \le \rho(v) \le \sqrt{h+1}\, \|v\|_2 .$$

The first inequality uses singleton segments; the second uses Cauchy–Schwarz along each segment. Norm squares add over different components, so J is an ℓ_2 sum of spaces isomorphic, component by component, to their coordinate ℓ_2 spaces. Its dual is the ℓ_2 sum of the component duals. Truncating first the components and then the coordinates proves finite-coordinate density in both J and X and proves reflexivity. The equivalence constants need not be uniform in height. Adding ancestors to a finite support produces a finite ancestral approximation head. The antichain norm in J is exactly ℓ_2, since each segment meets an antichain at most once. These elementary foundations rederive the particular source facts needed for the direct proof.

1.2 Source architecture and distinct imported inputs

The complete assigned scientific source was read, including all included modules. Its principal conclusion is $D^2 \geq 1 + k/4$ for X and for the infinite-height closed coordinate span $B_\infty \subset J^*_\infty$. The source's `metrics.tex`, lines 114–169, derives it from the companion arbitrary-displacement inequality

$$\|Q_A y\| \leq 2\sqrt{R^2 - \|p\|^2}, \qquad P_A p = p, \quad \|p \pm y\| \leq R. \tag{4}$$

Its own finite-head approximation theorem is required in each target. The full infinite-height dual is not substituted for B_∞. The midpoint-lens companion is the family interface also relevant to M638; its theorem is inherited here, not counted as a new M639 result. Its proof body was not freshly read for this assignment.

The assigned source has additional genuinely different arguments. `direct.tex`, lines 1–315, gives forest foundations, the fresh-root correction and a size-balancing descent. Lines 318–714 construct exact finite atomic laws and prediction-energy arguments, with endpoint or start counting. The terminal atom sampling is excluded even from the terminal binary-history sigma-algebra; the start argument uses a finite pair measure, not necessarily a probability measure. Lines 716–965 use imported path-cost energy estimates, with separate support and predictability requirements. The constants 495, 252, 400 and 1024 belong to those distinct inputs and cannot be interchanged.

`transfer-extensions.tex` treats squared, radial, power, parametric, quotient and qualitative losses. The two applications modules change the actual norms: linear versus quadratic path cost, prescribed segment starts versus incomparable segments, and recursive child exponents or joining-root conventions. The appendix records alternative forest midpoint inputs. This report strengthens selected deductions; it does not independently validate all companion intrinsic inequalities or a global Banach-space classification.

2 A shared fresh-subtree correction budget

The following refinement concerns the source's correction lemma, `direct.tex`, lines 127–175. Its construction is retained. The improvement is to account for a crossing and all exterior tests in the same subtree together.

Theorem 2.1 (Twelve instead of sixteen)**.** *Let A be finite ancestral, let $v, u \in c_{00}(T)$, and suppose $P_A v = v$, $Q_A u = u$. There is $c = Q_A c \in c_{00}(T)$ such that*

$$\rho(c) \leq \rho(u), \qquad \rho(v + c + \varepsilon u)^2 \leq \rho(v)^2 + 12\rho(u)^2 \quad (\varepsilon \in \{-1, 1\}). \tag{5}$$

The same c works for both signs.

Proof. The complement of A is a disjoint union of rooted subtrees U_d, with incomparable first exterior roots d. Put $u_d = u|_{U_d}$ and $a_d = \rho(u_d)$. Disjoint tests in these subtrees give

$$\sum_d a_d^2 = \rho(u)^2.$$

Only finitely many a_d are nonzero. When $a_d = 0$, set $c_d = 0$. If d has parent $p \in A$ and some suffix $[h, p]$ of the ancestor chain has $|\sum_{t \in [h,p]} v(t)| > 2a_d$, choose the shortest such suffix, denote its sum by P'_d, and put $c_d = -\operatorname{sign}(P'_d) a_d$. Otherwise put $c_d = 0$. For a component root with no parent in A also use zero. Set $c = \sum_d c_d e_d$. The antichain identity gives $\rho(c)^2 = \sum_d c_d^2 \leq \sum_d a_d^2$.

Fix one sign ε and any disjoint segment family testing $v + c + \varepsilon u$. At most one segment crosses from A into each U_d. If there is such a crossing, let P be its head sum, q its u_d tail sum, and w the

sum of the squares of the u_d sums on all other tested exterior segments in U_d. Disjointness gives the crucial shared budget

$$q^2 + w \le a_d^2. \tag{6}$$

The other exterior segments cannot contain d, so their c sums are zero. The entire contribution from this crossing and these exterior segments is

$$(P + c_d + \varepsilon q)^2 + w. \tag{7}$$

In particular, for any replacement head test E,

$$(P + c_d + \varepsilon q)^2 + w - E \le P^2 - E + 2Pc_d + c_d^2 + a_d^2 + 2|P + c_d|a_d. \tag{8}$$

If $|P| \le 2a_d$, choose $E = P^2$. The right side is at most $4a_d^2 + a_d^2 + a_d^2 + 6a_d^2 = 12a_d^2$. If $|P| > 2a_d$, a preferred suffix exists and lies within the head portion of this crossing, since that portion itself qualifies. When its sign agrees with P, $c_d = -\operatorname{sign}(P)a_d$, so $P(c_d + \varepsilon q) \le 0$. With $E = P^2$, the residual in (7) is at most $c_d^2 + 2|c_d q| + (q^2 + w) \le 4a_d^2$. When the signs disagree, the preferred suffix is properly inside the head portion. Splitting there produces two disjoint head segments whose squared test is

$$E = (P - P_d')^2 + (P_d')^2 \ge P^2 + 4|P|a_d.$$

Since $|c_d + \varepsilon q| \le 2a_d$, this absorbs the cross term; the remaining contribution is again at most $4a_d^2$ by (6).

If there is no crossing into U_d, all its exterior segments together have squared test at most $\rho(c_d e_d + \varepsilon u_d)^2 \le 4a_d^2$. All replacement head pieces, and the unchanged head segments, remain disjoint. Their total is at most $\rho(v)^2$. Summing the subtree budgets and taking the supremum proves (5). □

Remark 2.2 (The precise gain and its limit)**.** The source separately charges at most $12a_d^2$ for a crossing and $4a_d^2$ for exterior tests. Equation (6) prevents charging the same available u_d energy twice. The number twelve is attained in the relaxed scalar small-head budget by $a_d = 1$, $P = 2$, $c_d = q = 1$, $w = 0$, $\varepsilon = 1$, $E = P^2$. This shows sharpness of that scalar estimate under its declared constraints only. It does not construct a globally extremal forest test or prove the optimality of the correction theorem.

3 An adaptive witness and its initial budget

Pairings below are the actual duality $X \times J \to \mathbb{R}$, not a Euclidean replacement norm. A finite witness is a vector in $c_{00}(T)$ with its segment norm ρ.

Lemma 3.1 (One adaptive step)**.** *Let $x, x + z, x - z \in X$, let $v = P_A v \in c_{00}(T)$, and suppose $\|Q_A x\| \le \delta$. If $w = Q_A w \in c_{00}(T)$, $\rho(w) \le 1$, and $z(w) \ge \beta$, then for each $t \ge 0$ one of the two actual children $x' = x + \varepsilon z$ admits a witness $v' = v + c + \varepsilon t w$ satisfying*

$$x'(v') \ge x(v) + t(\beta - \delta), \qquad \rho(v')^2 \le \rho(v)^2 + 12t^2.$$

Here the correction c is supported outside A and has $\rho(c) \le t$.

Proof. Use Theorem 2.1 with $u = tw$. Averaging the two pairings with matched signs cancels the terms $x(tw)$ and $z(v + c)$:

$$\frac{1}{2}\sum_{\varepsilon=\pm 1}(x + \varepsilon z)(v + c + \varepsilon t w) = x(v) + x(c) + tz(w) \ge x(v) + t(\beta - \delta).$$

The last step uses $x(c) = (Q_A x)(c) \ge -\delta t$. At least one actual child attains this average. □

Theorem 3.2 (Finite, adaptively weighted descent)**.** *Consider k successive child selections of this kind, with every selected slope of norm at most R. Suppose a finite initial witness u_0 satisfies $\rho(u_0) \le 1$ and $x_0(u_0) \ge a_0 \ge 0$. At step j, after the current witness is known, choose A_j, w_j, z_j as in Lemma 3.1, with verified $g_j = \beta_j - \delta_j > 0$. Then*

$$a_0^2 + \frac{1}{12}\sum_{j=0}^{k-1} g_j^2 \le R^2. \tag{9}$$

The gains may depend on the preceding choices. No future branch or gain needs to be known when choosing a witness.

Proof. Start with $v_0 = a_0 u_0$. If $C_j = a_0^2 + \frac{1}{12}\sum_{i<j} g_i^2$, we maintain $x_j(v_j) \ge C_j$ and $\rho(v_j)^2 \le C_j$. At step j choose $t_j = g_j/12$ in the lemma. Both the guaranteed pairing increment $t_j g_j$ and the norm-square budget $12t_j^2$ equal $g_j^2/12$. This proves the induction. At the last step, $C_k \le x_k(v_k) \le R\sqrt{C_k}$. Division by $\sqrt{C_k}$ when positive proves the claim; the zero case is immediate. □

Corollary 3.3 (A stronger direct route)**.** *For $X = J^*$ defined in Section 1.1, every embedding of $\mathcal{D}_k$ with distortion D satisfies*

$$D^2 \ge 1 + \frac{k}{12}. \tag{10}$$

This proof does not use the companion midpoint inequality (4)*.*

Proof. Use the noncontracting normalization (1). Fix $0 < e < 1$. A finite primal norming approximation gives $x_0(u_0) \ge 1 - e$, $\rho(u_0) \le 1$. Such approximations follow from density in J and the definition of its dual norm.

At any stage, choose a finite ancestral A containing the current witness and satisfying $\|Q_A x\| \le \delta$, where $\delta + \eta + \nu \le e$ and all three numbers are positive. The infinitely many $P_A z_i$ lie in a bounded finite-dimensional set. Two of them have distance less than 2η. Since $\|z_i - z_l\| \ge 2$, their tails have distance greater than $2 - 2\eta$; one tail therefore has norm greater than $1 - \eta$. For that actual z_i, choose a finite pure-tail norming vector w with $\rho(w) \le 1$ and $z_i(w) > 1 - \eta - \nu$. Indeed, duality first gives a primal vector, and applying the contractive Q_A, taking finite approximations and normalizing preserves an arbitrarily small error. Thus $g_j \ge 1 - e$. Lemma 3.1 selects an actual child and a finite new witness. There are only k steps, so all these selections are legitimate. Theorem 3.2 yields $(1-e)^2(1+k/12) \le D^2$. Let $e \downarrow 0$. No norming vector, head approximation or stage supremum is required to attain its limit. □

The source's direct construction uses a fixed tail threshold and a fixed matched-sign gain, leading to $k \le 256D^2$. There are three separate changes here: the shared cost twelve, arbitrarily small finite-head losses at a fixed finite depth, and the initial norming budget a_0^2. None changes the graph exponent. The source's stronger $D^2 \ge 1 + k/4$ remains stronger; its proof uses the additional intrinsic midpoint theorem.

4 What finite branching can actually certify

The compactness selection above is qualitative in the head rank. A real finite-dimensional packing estimate makes its conditional finite version explicit.

Lemma 4.1 (Head packing)**.** *Let P have real rank d and let $\|Pz_i\| \le B$ for $1 \le i \le N$. If $\eta > 0$ and*

$$N > (1 + B/\eta)^d, \tag{11}$$

then distinct indices satisfy $\|Pz_i - Pz_l\| < 2\eta$. For rank zero, $N \ge 2$ suffices.

Proof. Otherwise the open radius-η balls around the Pz_i in the range of P are pairwise disjoint, and all lie in its radius-$(B+\eta)$ ball. Lebesgue volume in this real normed space then gives $N\eta^d \le (B+\eta)^d$, a contradiction. The argument does not require an inner-product norm or a preferred coordinate volume normalization. □

Corollary 4.2 (Conditional finite-branch certificate)**.** *Suppose that along the adaptive descent in X all slopes have norm at most R. At step j, assume the available deviations are $2\alpha_j$ separated, their projections on the selected rank-d_j ancestral head have norm at most B_j, and their number exceeds $(1+B_j/\eta_j)^{d_j}$. Suppose also that $\left\|Q_{A_j}x_j\right\| \le \delta_j$ and a finite tail norming vector is chosen with error at most ν_j. If $h_j = \alpha_j - \eta_j - \nu_j - \delta_j > 0$, then*

$$a_0^2 + \frac{1}{12}\sum_{j<k} h_j^2 \le R^2. \tag{12}$$

Proof. Packing provides a pair of close heads. Its tail half-difference has norm at least $\alpha_j - \eta_j$, so at least one actual deviation has tail norm at least that large. A finite pure-tail normer gives $\beta_j \ge \alpha_j - \eta_j - \nu_j$. Apply Theorem 3.2; larger actual gains only strengthen its conclusion. □

This is not a uniform finite-branch obstruction deduced from infinite branching alone. The ranks d_j can depend on the embedding, the current witness and the requested δ_j. The source supplies no uniform approximation-rate bound for those ranks. In upper-one normalization, $\alpha_j = 1/D$ and $B_j \le 1$; in noncontracting normalization, $\alpha_j = 1$ and $B_j \le D$. These two conventions must not be mixed.

Example 4.3 (An error budget with declared ranks)**.** Suppose the noncontracting upper bound is $R = 2$, every selected head has rank at most d, and each of η, ν, δ is at most $1/12$. More than 25^d available midpoints at each selected parent suffice for the packing step. Then $h_j \ge 3/4$. With an initial witness $a_0 \ge 3/4$, (12) gives $9/16 + 3k/64 \le 4$, hence $k \le 73$ for integer depth. If such ranks and initial data cannot be certified, this numerical obstruction cannot be asserted. In the countably branching limit the arbitrarily small losses instead give $k \le 36$ when $D = 2$.

5 Retaining the radius in a different cubic norm

This section changes both the space and the imported input. Join all finite-height components at a new root ρ. For a finitely supported array v, define the least nonnegative arrays p, q with $p_s - q_s = v_s$ and with each array dominating the Euclidean norm of its child values. Explicitly, if $a_s = (\sum_{t\in C(s)} p_t^2)^{1/2}$ and $b_s = (\sum_{t\in C(s)} q_t^2)^{1/2}$, then

$$p_s = \max\{a_s, v_s + b_s\}, \qquad q_s = \max\{a_s - v_s, b_s\}.$$

Require $v_\rho = 0$ and define

$$N_0(v) = p_\rho = q_\rho = \max\left\{\left(\sum_h p_{r_h}^2\right)^{1/2}, \left(\sum_h q_{r_h}^2\right)^{1/2}\right\}.$$

Write X_0 for its completion. On this root-zero subspace the free-root sum norm is $2N_0$, not N_0. In particular X_0 is not the forest dual X above. Ancestral projections containing ρ are contractive by the monotonicity of the recursion. Finite arrays are dense by completion, so such finite heads approximate each fixed vector; no uniform rank bound follows.

The assigned source imports from *Midpoint convexity from two recursive potentials*, Theorem 2.1, equation (2.4), the inequality

$$\frac{N_0(p+y)+N_0(p-y)}{2} \geq N_0(p)+G(N_0(p),N_0(y)), \qquad G(r,a)=\frac{a^3}{8(2r+a)^2}. \tag{13}$$

Here p is supported on a finite ancestral head containing ρ, and y is any nonzero completed tail vanishing on that head. The companion proof is not re-proved or independently audited here. All conclusions in this section are conditional on this exact norm-specific estimate. Replacing its coefficient or scaling the norm while keeping the same displayed radii would be invalid.

Theorem 5.1 (Retained-radius transfer)**.** *Assume* (13)*. For an embedding of $\mathcal{D}_k$ into X_0 with lower Lipschitz factor exactly one and upper factor D, let R_j be the supremum of the norms of the stage-j edge slopes. Then $1 \leq R_j \leq D$ and*

$$R_{j+1} \geq R_j + \frac{1}{8(2R_j+1)^2}. \tag{14}$$

In particular, if $b_0 = 1$ and $b_{j+1} = b_j + [8(2b_j+1)^2]^{-1}$, then $D \geq b_k$ and

$$k \leq \frac{4}{3}((2D+1)^3-27). \tag{15}$$

Proof. Fix an actual parent slope x of norm $r \geq 1$, and let $L = R_{j+1}$. For every $0 < \eta < 1$, choose a finite ancestral head and p supported on it with $N_0(x-p) \leq \eta$. The projected deviations are bounded in finite dimension. Choose distinct deviations z_i, z_l whose half-difference $w = (z_i - z_l)/2$ has $N_0(P_A w) \leq \eta$. All-pairs noncontraction gives $N_0(w) \geq 1$. Convexity of the two paired child slopes gives $N_0(x \pm w) \leq L$. Put $y = w - P_A w$. Then

$$N_0(y) \geq 1-\eta > 0, \qquad N_0(p \pm y) \leq L + 2\eta, \qquad |N_0(p) - r| \leq \eta.$$

This y need not be finitely supported, as permitted in the input. For $r \geq 0$, $a > 0$,

$$\partial_a G(r,a) = \frac{a^2(6r+a)}{8(2r+a)^3} > 0.$$

Consequently (13) implies $L + 2\eta \geq N_0(p) + G(N_0(p), 1-\eta)$. Let $\eta \downarrow 0$ for this fixed parent to obtain $L \geq r + [8(2r+1)^2]^{-1}$.

The function $\psi(r) = r + [8(2r+1)^2]^{-1}$ is continuous and strictly increasing for $r \geq 1$, since $\psi'(r) = 1 - [2(2r+1)^3]^{-1} > 0$. Take the supremum over the actual parents, using continuity even when it is not attained. This proves (14) and, inductively, $R_j \geq b_j$. Finally let $H(r) = \frac{4}{3}(2r+1)^3$. It is increasing and convex on $[1,\infty)$, and $H'(r) = 8(2r+1)^2$. Hence

$$H(R_{j+1}) - H(R_j) \geq H'(R_j)(R_{j+1} - R_j) \geq 1.$$

Sum the k inequalities and use $R_0 \geq 1$, $R_k \leq D$. □

Proposition 5.2 (A finite-error cubic certificate)**.** *In the preceding parent argument, suppose the actual verified errors are $N_0(x-p) \leq \zeta$ and $N_0(P_A w) \leq \eta$, while $N_0(w) \geq \alpha > \eta$. Then*

$$L \geq (r-\zeta)_+ + G((r-\zeta)_+, \alpha-\eta) - \zeta - \eta. \tag{16}$$

Proof. The input gives $L + \zeta + \eta \geq N_0(p) + G(N_0(p), \alpha - \eta)$. For $a > 0$, $\partial_r(r + G(r,a)) = 1 - a^3/[2(2r+a)^3] \geq 1/2$ on $r \geq 0$. Thus replacing $N_0(p)$ by its lower bound $(r-\zeta)_+$ preserves a valid lower estimate. □

This finite version retains explicit errors. The sharper limiting recurrence uses an arbitrarily small choice for every fixed parent; it does not assert that a preassigned finite computation finds vanishing errors uniformly across all parents.

6 Explicit comparisons and certificate boundaries

Example 6.1 (The cubic improvement at a fixed distortion)**.** For $D = 2$, (15) says $k \le 392/3$, hence $k \le 130$. The source's fixed-error argument gives $k \le 18432$ in the same norm N_0. More generally the new conditional distortion lower bound is

$$D \ge \frac{1}{2}\left((27 + 3k/4)^{1/3} - 1\right).$$

The recurrence in Theorem 5.1 can be stronger than this integrated estimate. Its first step is $b_1 = 73/72$. If $\phi(r) = [8(2r+1)^2]^{-1}$ and $A = 2r + 1$, direct expansion gives

$$H(r + \phi(r)) - H(r) = 1 + 16A\phi(r)^2 + \frac{32}{3}\phi(r)^3 > 1. \tag{17}$$

The integrated bound deliberately discards this positive remainder. There is no claim that embeddings attaining either estimate exist.

Target and proof route	Depth bound	Extra intrinsic input
X, source direct	$k \le 256D^2$	none beyond forest structure
X, present direct	$k \le 12(D^2 - 1)$	same structure
X, source midpoint	$k \le 4(D^2 - 1)$	companion midpoint theorem
X_0, source cubic	$k \le 4608D^2(D - 1)$	(13)
X_0, present cubic	$k \le \frac{4}{3}((2D + 1)^3 - 27)$	same cubic input

The table is a comparison of proved implications, not a ranking of the target spaces or a set of upper bounds on achievable distortion. Multiplying an entire norm by a constant does not change distortion; it does change the slope radii used when applying a displayed intrinsic inequality.

Example 6.2 (Why the ancestral hypothesis is substantive)**.** On a three-vertex chain, the array $(1, -1, 1)$ has $\rho^2 = 3$. Keeping only the first and last coordinates gives $(1, 0, 1)$, whose whole-chain test has value four, and in fact $\rho^2 = 4$. Thus an arbitrary coordinate deletion need not be contractive. Our head and tail projections are justified by ancestral geometry; they cannot be replaced by a generic list of observed coordinates.

6.1 A source-specific polynomial check and the Ram comparison

The original private chapter on between-knot constraints [6], read freshly in full, proves the Bernstein convex-hull rule and emphasizes that an exact polynomial part must be accompanied by a justified remainder bound. Its warning that checking finitely many samples is not a continuum certificate is directly relevant to the scalar relaxation used here, although its trajectory application is not a Banach-space theorem.

There is a small exact application to our correction proof. Normalize $a_d = 1$ in the small-head branch, absorb ε into q, and use $w \le 1 - q^2$. The required residual is bounded below by

$$F(P, c, q) = 11 - 2P(c + q) - c^2 - 2cq, \qquad |P| \le 2, \quad |c|, |q| \le 1.$$

Set $P = 4x - 2$, $c = 2y - 1$, $q = 2z - 1$. Writing $B_i^d(t) = \binom{d}{i} t^i (1 - t)^{d-i}$, the exact identity is

$$F(4x - 2, 2y - 1, 2z - 1) = \sum_{i=0}^{1}\sum_{j=0}^{2}\sum_{l=0}^{1} b_{ijl} B_i^1(x) B_j^2(y) B_l^1(z), \tag{18}$$

where the two matrices, with rows j and columns l, are

$$(b_{0jl}) = \begin{pmatrix} 0 & 12 \\ 8 & 16 \\ 12 & 16 \end{pmatrix}, \qquad (b_{1jl}) = \begin{pmatrix} 16 & 12 \\ 16 & 8 \\ 12 & 0 \end{pmatrix}.$$

Expansion verifies the identity. The basis products are nonnegative and sum to one on $[0,1]^3$, so the coefficients give $F \geq 0$ on the full declared box, including its zero corners. This is an alternative exact certificate for one scalar branch, not a new Bernstein theorem. The substantive forest step remains the proof of $q^2 + w \leq a_d^2$ and the disjoint replacement of head pieces. Without those statements the polynomial certificate says nothing about the original norm.

The freshly read portions of the later comparisons [7, 8] reinforce the same distinction between a finite reconstruction and control of what was omitted. They are comparison material, not substitutes for the original chapter or for this report's tree proofs. In particular, the core report's coercive elliptic residual theorem is not applied here: its differential operator, boundary conditions and coercivity assumptions do not describe the present segment norm. We do not claim a Green-operator or finite-Gram proof of diamond distortion.

7 Literature, limitations and research directions

The classical theory already connects countably branching diamonds with asymptotic midpoint geometry. Baudier and coauthors [2] prove quantitative and structural obstruction results; the converse described in their abstract is restricted to reflexive spaces with unconditional asymptotic structure, and is not an unrestricted characterization asserted here. Malthaner's metric approach [3] treats countably branching graph obstructions and power-type asymptotic midpoint convexity. Accordingly, compactness selection, volume packing, norm duality and scalar convex-potential integration are classical tools, not new general methods claimed in this report.

The public primary records were refreshed on October 9, 2026. The arXiv records checked were [2] v1, [3] v2, the May 2026 preprint [4], and the November 2025 book preprint [5]. The publisher record for [3] identifies its 2025 journal appearance; theorem numbering can differ from the preprint. These checks covered abstracts and version or publication metadata, not full external proofs. The recent tree-index preprint concerns broader ordinal and metric invariants; its existence does not validate an imported norm estimate in the assigned source. No exhaustive priority claim is made.

The chief benefit of the shared budget is that it works directly in the stated primal norm and explains precisely where the source's extra four units per subtree were spent. The adaptive witness retains actual norming gains, rather than requiring a uniform error chosen in advance. Its costs are finite but potentially large supports and a head-rank dependence for finite branching. Neither the proof nor the tests supply a measured computational speedup or an effective rank bound for arbitrary embeddings.

The cubic refinement has a different advantage: it avoids replacing the actual parent radius by a worst-case unit radius and avoids a fixed, unnecessarily large approximation loss. Its cost is dependence on the companion cubic theorem for the exact completed-tail class. It does not improve the cubic power of the obstruction, prove that input, or transfer the estimate to free-root or variable-exponent recursive norms.

Three focused extensions remain open within this analysis. First, a geometric correction using more than the present preferred suffix might improve twelve; scalar sharpness of the relaxed box does not settle that question. Second, quantitative coordinate approximation for a restricted embedding class could turn Corollary 4.2 into a genuinely uniform finite-branch result. Third, the exact positive remainder in (17) can sharpen a finite-depth scalar certificate without changing the intrinsic norm input. These are specific possible extensions, not announced completed theorems.

8 Verification and contribution scope

All twelve inventoried public text files were literally displayed in bounded chunks: the driver, preamble, bibliography, graph figure, README, and all seven scientific section files. Their total is 2,918 lines. This establishes a full reading of the assigned text, not an independent audit of every referenced companion theorem. Fresh private scientific reading comprised all 299 lines of the original clearance chapter and 240 lines of later comparison material. An additional 24 opening lines of the comparisons were read for file metadata. The owned ledger records exact intervals, unread complements, and hashes captured before and after the reads. No whole-corpus reading or fresh full companion-proof reading is claimed.

The owned executable checker uses exact integers and rational arithmetic. It independently computes small-tree segment norms by maximizing over all disjoint segment families through subset dynamic programming. It tests the actual preferred-suffix correction on chains and forks, the scalar cases, the adaptive square budget, cubic monotonicity and the exact potential remainder, worked examples, and the Bernstein polynomial identity. It excludes candidate tree fixtures with irrational subtree norms so that no rounding assertion is hidden. This is a declared fixture restriction, not a restriction in Theorem 2.1.

The final owned run passed 4,102 parameter cases and 13,880 exact assertions, including 2,975 full segment-norm correction cases. It excluded 2,024 candidate correction inputs whose subtree norms were not integers. No floating-point diagnostics, parent checks, earlier-report checks or foreign-checker runs are included. Finite tests supplement the proofs; they do not certify infinite branching, the imported cubic input or the source's global headline by themselves.

The contribution is C3 rather than a new global classification: the shared subtree budget and initialized adaptive descent improve a fully specified source proof, and the retained-radius transfer improves a distinct conditional source constant. The main source midpoint obstruction remains stronger in its own target, is credited as such, and is not counted twice through M638.

References

[1] OpenAI, *Distortion of countably branching diamonds from midpoint and tree energies*, September 27, 2026. Pinned public source tree. Full assigned text read; cited companion proof bodies remain imported.

[2] F. P. Baudier, R. M. Causey, S. J. Dilworth, D. Kutzarova, N. L. Randrianarivony, T. Schlumprecht and S. Zhang, *On the geometry of the countably branching diamond graphs*, J. Functional Analysis 273 (2017), 3150–3199. arXiv:1612.01984.

[3] R. Malthaner, *On Matoušek-like embedding obstructions of countably branching graphs*, J. Math. Anal. Appl. 543 (2025), 128896. arXiv:2310.03257v2.

[4] E. Basset, G. Lancien and A. Procházka, *On metric characterizations of tree and fragmentability indices of Banach spaces*, May 2026. arXiv:2605.28311v1.

[5] F. P. Baudier and G. Lancien, *Asymptotic and nonlinear geometries of Banach spaces and their interactions*, book preprint, November 2025. arXiv:2512.00817v1.

[6] M. Subbarao, *Between-knot constraints are continuous-time conditions*, original private RamCalBooks, volume 16, `chapters/ch05d-clearance.tex`, lines 1–299. Bernstein identity and hull proof: lines 66–105; subdivision and endpoint caution: lines 114–139; exact-part/remainder separation: lines 232–245. Local source only.

[7] M. Subbarao, private comparative report `res-math/TR-item-047-astra.tex`, scientific passages 162–199 and 346–391. Finite polynomial and residual-aware certificate comparison only.

[8] M. Subbarao, private comparative organizer *Measure-Aware Neural PDE Operators with Reconstructible Tokens and Residual Gates*, `TR-core-pat-4.tex`, scientific passages 233–325 and 637–699. Restricted reconstruction and full residual requirements only; no elliptic theorem is invoked for the tree norm.